TeraMove: Cascading Beam-to-Motion Adaptation for Blockage-Resilient Indoor THz Networks

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***Abstract*—We present TeraMove, a framework that adapts indoor THz Access Points (APs) to human blockage with a four-tier cascade. The cascade escalates from electronic beam steering to AP height changes and then to full 3D repositioning, and it moves an AP only when lighter adaptations do not provide threshold coverage. TeraMove uses spatial human-density context to detect THz link degradation from dynamic blockages. We describe how this context can be obtained from 3D LiDAR depth sensing; our evaluation uses a synthetic crowd-density model. TeraMove first establishes an optimal baseline layout for AP placement using a Mixed Integer Linear Programming (MILP) model over 3D coordinates and beam orientations. Simulations of an indoor floor with synthetic crowds show that crowd-induced blockages reduce coverage to about 45.6%. Beam steering and height changes add only 4.5 pp, because the people around a user block the links from the current APs. Full repositioning with a crowd-aware seed restores coverage to 99.0% but moves the APs by 83.4 m in total, while a standard PSO without the seed reaches only 54.0%.**

***Index Terms*—THz communication, human blockage, access point placement, beam steering, particle swarm optimization, mixed-integer linear programming**

I. Introduction

The emergence of 6G networks is driven by applications such as holographic communication, immersive extended reality, and digital twins that demand multi-Terabit-per-second data rates with ultra-low latency. The Terahertz (THz) band (0.1–10 THz), with far more available spectrum than sub-6 GHz and millimeter-wave (mmWave) bands, is widely regarded as an enabling physical layer for 6G. However, THz signals suffer extreme path loss and molecular absorption, requiring narrow, highly directional beams with unobstructed Line-of-Sight (LoS). In populated indoor spaces, a single human crossing the beam can cause attenuation exceeding 20 dB, potentially collapsing the link.

The straightforward solution is static placement optimization, fixing transmitting nodes at predetermined positions to maximize coverage. Existing methods typically treat this as a 2D problem at a uniform ceiling height. While adequate for an empty floor plan, such deployments assume the environment remains unchanged. However, human mobility continuously changes blockage conditions, and a layout providing near-perfect coverage can degrade as crowds shift.

3D LiDAR offers a reliable means to capture this changing context. Recent work has demonstrated accurate human tracking and counting indoors using 3D LiDAR [18], [19]. This spatial human-density information can characterize blockage conditions, while static over-provisioning can lead to unnecessary deployment costs.

We present *TeraMove*, a context-aware framework that treats transmitting node deployment as a continuous adaptation problem. It uses spatial human-density information to characterize dynamic blockage and first applies a Mixed-Integer Linear Programming (MILP) formulation to minimize the number of transmitting nodes while satisfying coverage and SINR constraints. When coverage falls below a threshold, a 4-tier HPSO framework adapts the network through beam steering, height adjustment, and, when necessary, full 3D repositioning. Under dense crowds Tiers 2 and 3 add little coverage, so most of the gain comes from Tier 4, while a Hungarian algorithm minimizes physical displacement when movement is required.

Although focused on THz, the framework is transmitter-agnostic and can also apply to mmWave small cells, Wi-Fi APs, and 5G base stations facing dynamic LoS blockages. We use the term "Access Points" (APs) throughout for generality.

The main **contributions** of this paper are:

- A 4-tier cascading architecture that escalates from beam steering to physical repositioning only when needed, improving coverage while minimizing node displacement.
- A MILP formulation that jointly optimizes 3D AP placement and azimuth orientation while minimizing the number of APs required to satisfy user-level SINR coverage.
- An HPSO algorithm for dynamic repositioning that prioritizes human-density-weighted coverage while penalizing unnecessary displacement.

II. Related Work

The THz band (0.1–10 THz) offers very large bandwidth for 6G, but suffers from high path loss, molecular absorption, and severe line-of-sight (LoS) blockage [1], [2], [5]. Human bodies can cause more than 20 dB attenuation [6], while pedestrian density and mobility strongly affect THz outage and SINR [11]. These characteristics make static deployments vulnerable to spatiotemporally varying human blockage.

Existing AP placement methods primarily optimize the network before deployment. Baswade *et al.* [3] formulated base-station placement using mixed-integer optimization, while Liu *et al.* [7] maximized mmWave LoS coverage with multiple APs. Such approaches provide effective initial layouts but do not adapt the deployment after propagation conditions change. PSO and its variants have also been applied to wireless optimization, including 3D UAV-BS placement [12] and mmWave

massive-MIMO beamforming [13]. However, these approaches mainly focus on optimizing placement or communication parameters, rather than combining electronic adaptation with physical AP reconfiguration when human blockage changes. Physical mobility offers another means of adapting wireless coverage as user locations vary. Wireless platforms based on UAV can adjust their position in three dimensions to improve coverage and optimize their trajectories [12], [14]. Similarly, movable APs have been studied for indoor WLANs, where the AP locations are changed according to user distribution [15]. Movable antennas further exploit spatial channel variations by dynamically changing antenna positions [16]. These approaches do not jointly consider indoor THz beam adaptation, vertical AP adjustment, and progressive full-3D repositioning based on human blockage.

Dynamic adaptation also requires information about human movement. Vision-based learning has been used to predict impending mmWave/THz blockages [17]. Specifically, 3D LiDAR point clouds have been used for high-resolution indoor crowd counting and geometry tracking [18], [19]. These sensing approaches can provide the spatial human-density information required for blockage-aware adaptation.

III. System Model and Problem Formulation

The indoor environment is discretized into a 3D volumetric grid. Let $\mathcal{U}$ denote the set of target user equipment locations requiring coverage, and let $\mathcal{A}$ denote the set of candidate AP locations. Let the 3D coordinates of the user device u be (x_u, y_u, z_u) and the candidate AP a be (x_a, y_a, z_a). The volumetric Euclidean distance $d_{u,a}$ is calculated as:

$$d_{u,a} = \sqrt{(x_u - x_a)^2 + (y_u - y_a)^2 + (z_u - z_a)^2} \quad (1)$$

where the user height is fixed at $z_u = 1.0$ m, and the optimization algorithm selects an optimal AP mounting elevation z_a from a discrete set of candidate heights. The floor is divided into $1\text{ m} \times 1\text{ m}$ cells, and each user is placed at the centre of its cell. This channel model serves as the evaluation setting; our contribution is the adaptation mechanism of Section IV, which does not depend on the details of the model.

The base Free Space Loss (FSL) for a signal operating at frequency f over distance $d_{u,a}$ is derived from the Friis transmission equation:

$$PL_{\text{FSL}}(d_{u,a}) = 20\log_{10}\left(\frac{4\pi f d_{u,a}}{c}\right) \quad (2)$$

where c is the speed of light in vacuum. The total deterministic (line-of-sight) path loss $PL_{\text{LoS}}(u,a)$ for THz link further incorporates molecular absorption loss and static attenuation [1]:

$$PL_{\text{LoS}}(u,a) = PL_{\text{FSL}}(d_{u,a}) + PL_{\text{ABS}}(d_{u,a}) + L_{\text{static}} \quad (3)$$

where $PL_{\text{FSL}}(d_{u,a})$ represents the free-space path loss, L_{static} denotes the attenuation penalty from static infrastructure, and $PL_{\text{ABS}}(d_{u,a})$ accounts for atmospheric molecular absorption at high frequencies [2], calculated in dB as:

$$PL_{\text{ABS}}(d_{u,a}) = 4.343 \cdot k(f) \cdot d_{u,a} \quad (4)$$

with $k(f)$ representing the frequency-dependent molecular absorption coefficient. At 300 GHz in indoor air (23 °C, 10 g/m^3 water vapour), ITU-R P.676 gives 6.68 dB/km, i.e., $k(f) = 1.54 \times 10^{-3}$ m^{-1} [4].

We model two distinct types of blockages: static and dynamic. For static obstacles, let $\mathcal{W}$ denote the set of walls within the environment. If the 3D line segment connecting u and a intersects a wall $w \in \mathcal{W}$, a material-specific penetration penalty L_w (e.g., concrete, wood, glass) is accumulated:

$$L_{\text{static}} = \sum_{w \in \mathcal{W}_{\text{intersect}}} L_w \quad (5)$$

Furthermore, THz signals suffer severe attenuation when occluded by human bodies [6]. We define an expected dynamic loss based on the normalized human density score $D_h(c) \in [0,1]$ for each grid cell (computed in Section IV-A) over the set of grid cells $\mathcal{C}_{u,a}$ intersected by the 2D projection of the link, excluding the cells of the AP and of the user. A human only obstructs the signal if the elevation of the beam $z(c)$ as it passes through cell c drops below the standard human height h_{human}. Using linear interpolation based on the 2D distance $d^{2D}_{u,c}$ from the user to the point of cell c closest to the user and the total 2D distance $d^{2D}_{u,a}$, the lowest beam elevation in cell c is given as:

$$z(c) = z_u + \frac{d^{2D}_{u,c}}{d^{2D}_{u,a}}(z_a - z_u) \quad (6)$$

The expected dynamic blockage loss is then defined using an indicator function $\mathbb{I}(\cdot)$, which equals 1 if the beam strikes the human body and 0 otherwise:

$$L_{\text{dynamic}} = L_h\Big[1 - \prod_{c \in \mathcal{C}_{u,a}} \Big(1 - D_h(c)\,\mathbb{I}\big(z(c) \le h_{\text{human}}\big)\Big)\Big] \quad (7)$$

where L_h is the empirical penalty caused by human blockage. The product is the probability that no cell on the path blocks the link, so L_{dynamic} never exceeds L_h.

We adopt a flat-top sector antenna model parameterized by the half-power beamwidth (HPBW), θ_{HPBW}, and the AP's mounting azimuth orientation ϕ_a. Since APs are mounted above users ($z_a > z_u$) with optimal downward tilt, signal gain is calculated only in the horizontal plane.

Let $\phi_{u,a}$ denote the spatial angle of user u relative to AP a. The shortest angular deviation $\Delta\phi$ between the user's location and the AP's orientation, is calculated as:

$$\Delta\phi = \left|(\phi_{u,a} - \phi_a + 180^\circ) \bmod 360^\circ - 180^\circ\right| \quad (8)$$

The transmit gain $G_{\text{tx}}(\Delta\phi)$ is:

$$G_{\text{tx}}(\Delta\phi) = \begin{cases} G_{\max}, & \text{if } \Delta\phi \le \frac{\theta_{\text{HPBW}}}{2} \\ G_{\min}, & \text{otherwise} \end{cases} \quad (9)$$

where $G_{\min}$ is practically negligible. A user directly below the AP is treated as inside the main lobe. The main-lobe gain depends on the beamwidth as $G_{\max} = 10\log_{10}\big(K/(\theta_{\text{HPBW}}\,\theta_{\text{el}})\big)$. The elevation beamwidth θ_{el} is fixed, and K is set so that a 60° beam has 30 dBi:

$$G_{\max}(\theta_{\text{HPBW}}) = 30 + 10\log_{10}\left(\frac{60^\circ}{\theta_{\text{HPBW}}}\right) \text{ dBi} \quad (10)$$

which gives 33.0 dBi at 30° and 27.0 dBi at 120°. The total path loss $PL(u,a)$ combines the deterministic LoS component with the expected dynamic human blockage:

$$PL(u,a) = PL_{\text{LoS}}(u,a) + L_{\text{dynamic}}(u,a) \tag{11}$$

Finally, the received power $P_{\text{rx}}^{(u,a)}$ (in W) is obtained from the total path loss and the antenna gain as:

$$P_{\text{rx}}^{(u,a)} = 10^{\frac{P_{\text{tx}}+G_{\text{tx}}(\Delta\phi)+G_{\text{rx}}-PL(u,a)-30}{10}} \tag{12}$$

where P_{tx} is transmit power in dBm and G_{rx} is receive antenna gain. User u is served by AP $a^*(u)$, and its SINR is

$$\text{SINR}_u = \frac{P_{\text{rx}}^{(u,a^*)}}{N_0 + \sum_{b\neq a^*} P_{\text{rx}}^{(u,b)}} \tag{13}$$

The coverage is the density-weighted share of cells that meet the SINR threshold:

$$\text{Cov} = \frac{\sum_u D_h(u)\,\mathbb{I}(\text{SINR}_u \geq \text{SINR}_{\text{th}})}{\sum_u D_h(u)} \tag{14}$$

On an empty floor ($D_h = 0$ everywhere), Cov is the fraction of cells that meet the threshold.

IV. Proposed Work

Our objective is to ensure that every user within the indoor environment receives a minimum required SINR (SINR_{th}) using the fewest THz APs possible, even as human crowds dynamically reshape the blockage landscape. We address this in two stages: first, we formulate the static baseline placement as a MILP to compute an optimal initial deployment over a discretized candidate set; second, we introduce an HPSO-based dynamic repositioning framework with a 4-tier cascading architecture to adapt the deployment in real time when crowd shifts degrade user coverage.

A. Context Acquisition from Human-Density Sensing

TeraMove uses spatial human-density information to capture changes in the blockage environment. In practice, this contextual data is obtained using 3D LiDAR sensors, which generate real-time topographical point clouds of the room. By subtracting the static background, we isolate dynamic points corresponding to humans or moving furniture. The human density score $D_h(c) \in [0,1]$ for a specific vertical column is calculated as:

$$D_h(c) = \min\left(1, \frac{N_{\text{dyn}}(c)}{N_{\text{max}}}\right) \tag{15}$$

where $N_{\text{dyn}}(c)$ is the number of dynamic LiDAR points detected in the vertical column at cell c, and N_{max} is the expected maximum number of points for a single human body. In our evaluation, we generate the equivalent human-density map with a synthetic crowd model (Section V) to evaluate coverage and trigger AP adaptation when needed.

B. Static Baseline Placement via MILP

We formulate the baseline optimization problem as a Mixed-Integer Linear Programming (MILP) model. The primary objective is to minimize the total number of APs deployed, expressed by Eq. (16):

$$\min \sum_{a\in\mathcal{A}} W_a \tag{16}$$

where $W_a \in \{0,1\}$ equals 1 if an AP is deployed at location a, and 0 otherwise. Let $Y_{u,a} \in \{0,1\}$ denote whether user u is associated with AP a. Eq. (17) ensures that each target sub-region is associated with exactly one deployed AP. Eq. (18) mandates that a user u can only logically associate with a candidate location a if an AP is physically deployed there.

$$\sum_{a\in\mathcal{A}} Y_{u,a} = 1, \quad \forall u \in \mathcal{U} \tag{17}$$

$$Y_{u,a} \leq W_a, \quad \forall u \in \mathcal{U}, \forall a \in \mathcal{A} \tag{18}$$

In practical deployment scenarios where satisfying 100% coverage is physically or economically infeasible, Eq. (17) is relaxed to an inequality ($\sum_{a\in\mathcal{A}} Y_{u,a} \leq 1$), and a global coverage constraint is introduced. Let $\eta \in (0,1]$ denote the required proportion that must be served. To strictly enforce this spatial coverage ratio, the total number of successfully associated sub-regions must satisfy the following constraint:

$$\sum_{u\in\mathcal{U}} \sum_{a\in\mathcal{A}} Y_{u,a} \geq \eta|\mathcal{U}| \tag{19}$$

The next constraint enforces the minimum SINR in each sub-region. For every user–AP pair, the received signal plus a big-M relaxation term must exceed the SINR threshold scaled by the total interference-plus-noise:

$$P_{\text{rx}}^{(u,a)} + \text{Inf}\cdot(1 - Y_{u,a}) \geq \text{SINR}_{\text{th}} \cdot \left(N_0 + \sum_{\substack{b\in\mathcal{A}\\ b\neq a}} P_{\text{rx}}^{(u,b)} W_b\right), \quad \forall u \in \mathcal{U}, a \in \mathcal{A} \tag{20}$$

When $Y_{u,a} = 1$ the big-M term vanishes and the constraint enforces the SINR requirement; when $Y_{u,a} = 0$ the large constant *Inf* makes the inequality trivially satisfied, deactivating it for unassociated links [3]. Because $P_{\text{rx}}^{(u,b)}$ is precomputed for each (u,b) pair and W_b is binary, the entire formulation remains linear and solvable by standard MILP solvers.

C. Dynamic AP Repositioning via HPSO

While MILP optimizes static deployments, indoor crowds are dynamic. To avoid static over-provisioning, we dynamically reposition APs to track evolving human densities.

We achieve this using a Hierarchical Particle Swarm Optimization (HPSO) algorithm based on [8], [10], which activates search dimensions in tiers. Each particle updates its position x_i (encoding AP parameters) and velocity v_i over T iterations:

$$v_i^{(t+1)} = w\cdot v_i^{(t)} + c_1 r_1 (p_i^{\text{best}} - x_i^{(t)}) + c_2 r_2 (g^{\text{best}} - x_i^{(t)}) \tag{21}$$

where inertia weight w decreases linearly (0.9 to 0.4); $c_1 = c_2 = 1.5$; $r_1, r_2 \sim U(0,1)$; and p_i^{best} and g^{best} denote personal and global bests. Positions update as $x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)}$ with velocity clamping.

The swarm's composite fitness function is:

$$F = w_c \cdot R_{\text{cell}} + w_d \cdot R_{\text{density}} + B - \beta \cdot M - \gamma \cdot C_p \quad (22)$$

where R_{cell} is the proportion of cells meeting the SINR threshold, R_{density} is the covered human density ratio (the coverage defined in Section III), $B = 0.5$ is a target-coverage bonus added when $R_{\text{density}} \geq 0.9$ ($B = 0$ otherwise), and M penalizes physical displacement:

$$M = \frac{1}{n\sqrt{L^2 + W^2 + H^2}} \sum_{i=1}^{n} \|r_i - r_i^{\text{prev}}\| \quad (23)$$

where r_i^{prev} is the position of AP i before adaptation and $L \times W \times H$ is the building size. Note that D_h has two roles: it is the occlusion probability in (7) and the demand weight in R_{density}. C_p prevents APs from clustering too closely:

$$C_p = \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} \max\left(0,\ 1 - \frac{\|r_i - r_j\|}{d_{\min}}\right) \quad (24)$$

where n is the AP count, $\|r_i - r_j\|$ is the horizontal (2D) distance between APs i and j, and $d_{\min}$ is the minimum inter-AP spacing (Table I). We prioritize human-centric coverage ($w_c = 0.4, w_d = 0.6$) over displacement ($\beta = 0.25$) and clustering ($\gamma = 0.3$) penalties. The resulting small physical displacements are largely due to the framework's strict tier-clamping architecture prioritizing zero-cost electronic steering over physical motion, rather than the β magnitude alone.

1) 4-Tier Cascading Optimization Architecture: Since moving APs incurs service interruption and mechanical wear, we introduce a 4-tier cascading optimization architecture that progressively unlocks more degrees of freedom only when simpler adjustments prove insufficient.

- **Tier 1 (Baseline Evaluation):** If current density-weighted coverage meets the target ($\geq 90\%$), no action is taken, avoiding unnecessary computation and movement.
- **Tier 2 (Beam Parameter Optimization):** HPSO first optimizes only the azimuth (ϕ_a) and beamwidth (θ_{HPBW}). These electronic adjustments do not move any AP.
- **Tier 3 (Height Adjustment):** If Tier 2 coverage remains insufficient, the optimizer adjusts the mounting height (z_a), which needs a minor Z-axis movement.
- **Tier 4 (Full Spatial Repositioning):** As a last resort, horizontal coordinates (x_a, y_a) are unlocked for complete 3D relocation. To limit displacement, one-third of the swarm starts within ± 2 m of the current APs. One particle starts from a crowd-aware greedy layout: APs are added one at a time from a candidate set (every cell centre, heights 2.0, 2.6 and 3.0 m, all azimuths, beamwidths 60°, 90° and 120°), each time the AP that adds the most density-weighted coverage under (13). The Hungarian step (Section IV-C2) matches it to the current APs.

At each tier, the HPSO locks the velocity components of the dimensions not yet activated. One particle starts at the current configuration, so a tier never returns a configuration with lower fitness than its input.

2) Displacement Minimization via Hungarian Algorithm: When the HPSO produces an optimized set of AP positions, we use the Hungarian algorithm [9] to find the assignment of existing APs to new positions that minimizes aggregate physical relocation distance.

Let $\mathcal{R}_{\text{prev}} = \{r_1, \ldots, r_n\}$ denote the n current AP positions and $\mathcal{R}_{\text{new}} = \{r'_1, \ldots, r'_n\}$ denote the n optimized positions. We construct a cost matrix $C \in \mathbb{R}^{n \times n}$ where each entry C_{ij} represents the displacement required to move existing AP r_i to new position r'_j:

$$C_{ij} = \sqrt{(x_i - x'_j)^2 + (y_i - y'_j)^2 + (z_i - z'_j)^2} \quad (25)$$

The Hungarian algorithm finds the one-to-one assignment with the least total displacement, $\sigma^* : \{1, \ldots, n\} \to \{1, \ldots, n\}$:

$$\sigma^* = \arg\min_{\sigma} \sum_{i=1}^{n} C_{i,\sigma(i)} \quad (26)$$

V. Experimental Setup and Simulation Results

For the experimental setup, we considered IIT Bhilai Engineering department layout. A synthetic crowd model generates the human density. A crowd has 9 circular clusters of radius 2 m with random centres, and the density falls linearly from 1 at a cluster centre to 0 at its edge. Each run and each time step draws a new crowd, and every result is the mean over 30 runs with independent random seeds. The simulation parameters are summarized in Table I.

We set $\text{SINR}_{\text{th}} = 5$ dB as evaluation criterion, the minimum signal quality for reliable THz link establishment. At 300 GHz, the short wavelength makes signals susceptible to blockage; a 5 dB threshold ensures sufficient power for basic connectivity after accounting for molecular absorption, wall penetration losses, and co-channel interference, consistent with the minimum SINR for low-order modulation in THz feasibility studies [5].

A. Static Baseline Deployment

We first evaluate the MILP formulation in a static environment using the Gurobi optimizer. Solving to a 0% optimality gap, the solver identifies that 14 APs are needed for 100% coverage at $\text{SINR}_{\text{th}} = 5$ dB. By evaluating candidate elevations in 0.2 m increments from 2.0 m to 3.0 m, the optimization enables APs to transmit above 1.0 m cubicles, yielding a 20% higher peak SINR compared to APs at a uniform height. Fig. 1 confirms the required SINR is met across all grid cells.

Raising SINR_{th} from 1 dB to 5 dB increases the AP count from 9 to 10 for 90% coverage. Wider beamwidths (90°–120°) require only 10 APs for 90% coverage, whereas narrow beams (60°–75°) require 15, as broader beams yield better overall indoor coverage.

TABLE I: Simulation Parameters

Parameter	Value
Environment Dimensions	$32\text{ m} \times 20\text{ m} \times 3\text{ m}$
Operating Frequency (f)	300 GHz
Transmit Power (P_{tx})	20 dBm
Main-Lobe Gain ($G_{\max}$)	$30 + 10\log_{10}(60^\circ/\theta_{\text{HPBW}})$ dBi
Receive Gain (G_{rx})	0 dBi
Side-Lobe Leakage ($G_{\min}$)	Negligible ($\leq$ Noise Floor)
Evaluated Beamwidths (θ_{HPBW})	$30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ, 120^\circ$
Azimuth Rotation Increments	45° ($\pi/4$ rad, from North)
Evaluated Elevations (z_a)	2.0 m to 3.0 m (0.2 m step)
SINR Threshold (SINR_{th})	5 dB
Target Coverage Threshold (η)	0.90 (90%), 0.95 (95%), 1.0 (100%)
System Noise Floor (N_0)	-64 dBm (-174 dBm/Hz, 10 GHz, 10 dB NF)
Static Blockage Penalty (L_w)	Solid: 30 dB, Wood: 20 dB, Glass: 15 dB
Static Cubicle/Desk Height	1.0 m
Human Blocker Penalty (L_h)	20 dB (height 1.6 m)
Molecular Absorption ($k(f)$)	1.54×10^{-3} m^{-1} (6.68 dB/km)
Grid Cell, User Height (z_u)	$1\text{ m} \times 1\text{ m}$, 1.0 m
Min. AP Spacing ($d_{\min}$)	3 m
Fitness ($w_c, w_d, \beta, \gamma, B$)	$0.4, 0.6, 0.25, 0.3, 0.5$
Swarm Size, Iterations per Tier (S, T)	$50, 20$
AP Actuator Speed	0.2 m/s

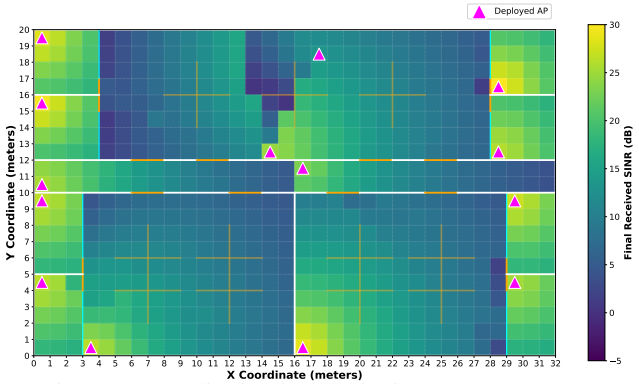


Fig. 1: SINR distribution achieving 100% coverage.

B. Tier-wise Coverage and Displacement Analysis

The dynamic evaluation uses the channel model of Section III. Under this model the 14 MILP APs cover 60.9% of the empty floor, so we first re-tune their heights and beams once on the empty floor, keeping their positions. This gives 98.0% coverage and places all APs at 2.0 m, the best height when no one blocks the links.

We tested the 4-tier cascading HPSO under human blockage conditions that drop initial coverage to 45.6% (mean over 30 crowds; all tiers are run on every crowd). Table II summarizes each tier's contribution, with 95% confidence intervals. From a 14-AP baseline, Tier 2 recovers 2.8 pp (to 48.4%) without any physical movement (0.0 m). Tier 3 adds 1.7 pp for 0.9 m of displacement, and Tier 4 contributes a final 48.9 pp at 83.4 m in total, for a coverage of 99.0%. Tier 4 meets the 90% target on 100.0% of the crowds. Without the body loss, the same layout would cover 98.1% of these crowds, so the outages come from blockage. By (6), the ray is below head height on the part of the link within $(h_{\text{human}} - z_u)/(z_a - z_u)$ of its length from the user, i.e., the last 60% for $z_a = 2.0$ m and 30% for $z_a = 3.0$ m. The people around a user inside a crowd therefore block the user from the current APs, and only APs moved close to the crowd (Tier 4) restore the link.

All coverage values assume best-server re-association: each user is served by the strongest AP for the current crowd (a^* in (13)). This evaluation-time association is separate from the design-time association $Y_{u,a}$ of the MILP. Section V-D compares it with a layout that keeps the design-time association.

TABLE II: Tier-wise Coverage and Displacement Summary (mean $\pm$ 95% CI over 30 crowds; gain in pp in brackets)

Tier	Coverage (%)	Displ. (m)	Actuation (s)	Adjustment Type
Baseline	45.6 ± 3.4	0.0	0	None
Tier 2	48.4 ± 3.0 (+2.8)	0.0	0	Electronic (beam steering)
Tier 3	50.1 ± 3.2 (+1.7)	0.9	2.3	Minor physical (Z-axis)
Tier 4	99.0 ± 0.4 (+48.9)	83.4	73.7	Full 3D repositioning

The 4-tier approach prioritizes the least disruptive adjustments. Tier 1 retains the existing configuration when coverage is sufficient, conserving computational and mechanical resources. Tier 2 gives 5.2% of the total improvement at zero physical cost by electronically adjusting beam direction and width via phased-array controls. The actuation time in Table II is the largest single-AP displacement divided by the 0.2 m/s actuator speed, since APs move in parallel. The optimizer takes about 3.5 s for Tier 2 and 56.5 s for Tier 4 on one CPU core.

Tier 3 is preferred over Tier 4 as minor vertical adjustments consume less time and energy than horizontal robotic relocation. Tier 4 is reserved for when blockage severity renders current AP coordinates entirely unviable. For physical movements in Tiers 3 and 4, the Hungarian assignment algorithm (Section IV-C2) replaces the nearest-neighbor approach. This reduces the Tier-4 displacement from 104.9 m to 83.4 m, a 20.5% saving (Wilcoxon $p < 0.001$).

C. Temporal Crowd Adaptation

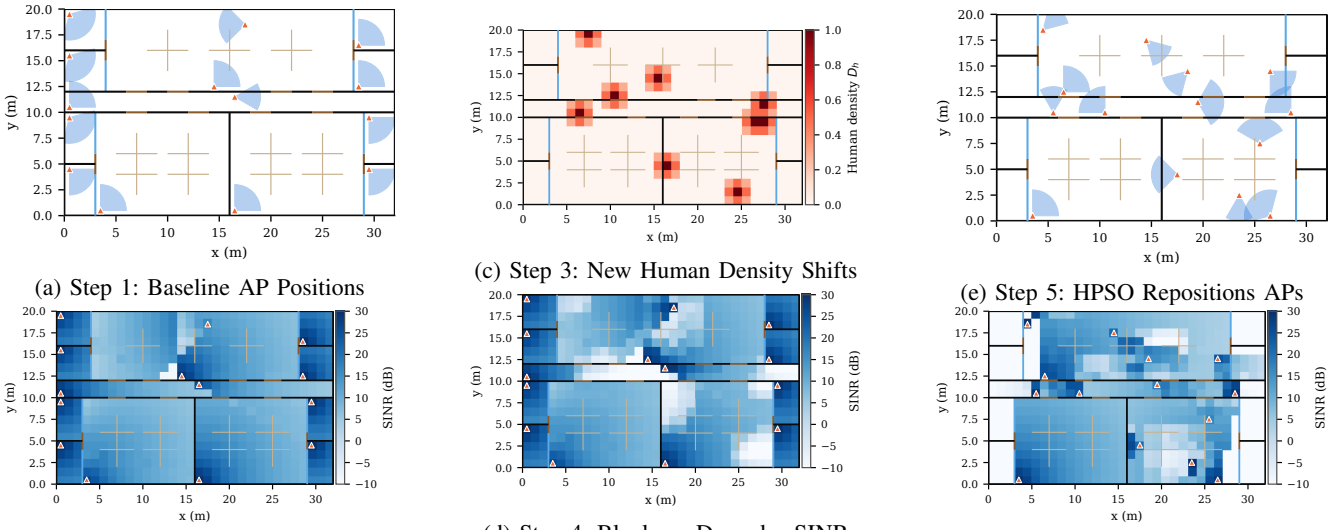
To evaluate robustness under changing crowds, we simulated three consecutive crowd shifts at discrete time steps (t_0, t_1, t_2), each drawn independently from the synthetic crowd model. Each step starts from the layout left by the previous step. Before adaptation, coverage is 44.1% at t_0, 28.5% at t_1 and 27.1% at t_2.

HPSO raises coverage from 44.1% to 98.9% at t_0, from 28.5% to 99.0% at t_1, and from 27.1% to 98.7% at t_2. The total AP displacement is 87.4 m, 89.2 m and 88.3 m at the three steps. Tiers 2 and 3 do not reach the 90% target, so the cascade reaches Tier 4 at every step and meets the target in 100.0% of the steps. Coverage before adaptation is lower at t_1 and t_2, since the layout from the previous step was placed for a different crowd.

Fig. 2 presents the adaptation sequence at t_0 for one run: from the commissioned AP configuration (a) and its SINR on the empty floor (b), a new crowd shift (c) lowers coverage to 46.6% (d), and HPSO re-angles and moves the APs by 75.2 m in total (e), which raises coverage to 97.6% (f).

D. Comparison with Alternative Approaches

We evaluate HPSO against standard Particle Swarm Optimization (PSO) and a Genetic Algorithm (GA). Combinatorial



(a) Step 1: Baseline AP Positions

(b) Step 2: Established Stable SINR

(c) Step 3: New Human Density Shifts

(d) Step 4: Blockage Degrades SINR

(e) Step 5: HPSO Repositions APs

(f) Step 6: SINR After Adaptation

Fig. 2: Adaptation sequence at t_0: The commissioned AP layout (a) yields stable SINR (b). A crowd shift (c) degrades signals (d), triggering HPSO to reposition and re-angle APs (e), which raises coverage from 46.6% to 97.6% (f).

approaches such as Greedy Set Cover are excluded because they discard MILP-derived AP positional states, treating each time step independently and failing to preserve the physical movement continuity required for mobile AP deployments. The unconstrained PSO serves as a baseline to isolate the benefits of hierarchical tier constraints; it optimizes the identical composite objective function (22) (including displacement penalty βM) but without tier-clamping, updating all 3D coordinates and beam parameters simultaneously. GA serves as a population-based benchmark to verify improvements are not PSO-specific. Simulations use swarm size $S = 50$ and $T = 20$ iterations per tier; PSO and GA get $3T$ iterations, the same worst-case budget as Tiers 2–4. We add two reference arms: the static layout with design-time association, and re-association plus Tier 2 only. PSO and GA get no seed.

Fig. 3 shows coverage after one adaptation step as human density increases. Coverage falls as the crowd grows for the static arms and the unseeded baselines, while HPSO stays above 95%. Re-association alone (not shown) adds 0.1 pp over the design-time association, because for 97.1% of the crowd the design-time AP is still the strongest one. HPSO reaches 97.7% on average, against 47.2% for the static layout, and Tier 2 alone reaches 51.0%. HPSO meets the 90% target in 99.2% of the events. Standard PSO and GA reach 57.8% and 59.7% (Wilcoxon $p < 0.001$ and $p < 0.001$ against HPSO).

The hierarchical structure tries electronic beam steering before physical relocation, but under these crowds Tier 4 is needed in every event. Over the three crowd shifts of Section V-C, HPSO moves the APs by 264.9 m in total, against 31.2 m for standard PSO and 80.0 m for GA; the extra movement raises mean coverage to 98.9%, against 54.0% and 56.1%. In those runs (9 clusters) the HPSO result equals its greedy seed in every step. In the single-step events of Fig. 3, HPSO moves the APs by 102.3 m on average, against 7.6 m

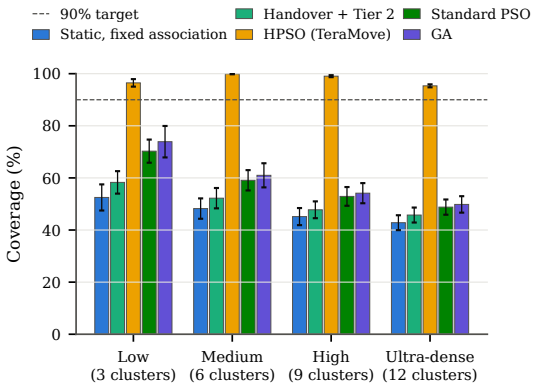


Fig. 3: Coverage after one adaptation step for HPSO, standard PSO, GA and two reference arms (mean over 30 crowds per level; error bars are 95% CIs; dashed line: 90% target).

for PSO and 18.1 m for GA. At the lower crowd levels the swarm finds layouts with higher fitness than the seed: slightly less coverage, but less movement.

VI. Discussion

The results presented in this work show how far the 4-tier cascading HPSO framework can recover coverage, and the practical deployment of this system can be realized through two distinct operational paradigms: real-time reactive adaptation or scheduled time-based adaptation.

In a **real-time adaptation model**, HPSO continuously updates the human-density spatial matrix using crowd-sensing data such as LiDAR, and is activated when coverage falls below 90%. Tier 2 first handles the blockage using electronic beam steering. If this is not enough, Tiers 3 and 4 adjust the AP height and perform full 3D repositioning. Alternatively, a **time-based scheduled system** can use historical footfall data to estimate density patterns and pre-compute

AP configurations for specific times, avoiding the need for continuous sensing. For physical movement, Tier 3 can use linear actuators, while Tier 4 can use ceiling-mounted robotic rails. Flexible PoE cable management keeps the APs connected during movement. The Hungarian algorithm minimizes the total AP displacement in each reconfiguration cycle.

Finally, practical deployments must account for actuation latency (rails moving at 0.2 m/s need about 73.7 s for the largest single-AP move in Table II) and the upfront hardware cost of robotic infrastructure compared to denser static over-provisioning. In our results the coverage target in a crowd is met only by moving APs close to the crowd: Tier 4 moves the APs by 83.4 m in total, against 0.9 m for Tiers 2 and 3.

Limitations. The blockage model treats each crossed cell as an independent blocker with probability $D_h(c)$ and a fixed loss L_h; measured body-loss statistics could change the results. Re-association is ideal: a user moves to the strongest AP at no cost, and handover delay is not modelled. The channel is LoS only, so reflected paths that could serve blocked users are ignored. The crowd density comes from a synthetic cluster model, not from measured LiDAR data, and the optimizer sees the exact density map; errors in the density estimate would affect both the blockage model and the fitness. PSO and GA start without the greedy seed. HPSO returns the seed itself in 85.0% of the Fig. 3 events and in every temporal step, so most of the Tier-4 gain comes from the seed. Finally, the gain model in (10) keeps the elevation beamwidth fixed and is anchored at 30 dBi for a 60° beam.

VII. Conclusion

This paper presents a framework for 6G Terahertz (THz) Access Point deployment in indoor spaces with moving crowds. We initially demonstrated that optimizing AP height and beam orientation via a 3D MILP formulation reduces the required number of APs. However, because static layouts degrade when moving human crowds block high-frequency signals, we introduced a human density metric to track blockage in each cell.

To overcome these blockages, we developed a 4-tier Hierarchical Particle Swarm Optimization (HPSO) architecture. By escalating from zero-movement electronic beam steering to full 3D robotic repositioning only when lighter tiers fall short, the framework restores coverage under crowd blockage. Beam steering and height changes add little in dense crowds, because the people around a user block the links from the current APs. Full repositioning with a crowd-aware seed reaches 98.9% coverage over three crowd shifts, but it moves the APs by 264.9 m in total. The Hungarian algorithm then matches the APs to the new positions with the least total movement.

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